

SAMBP Technical Report TR-31/2026

Zero-Sequence Compensation for IBR Earth Faults (21G)

PhD Thesis Supporting Report — Distance Relay Series III

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Abstract

This report analyses the zero-sequence compensation factor k_0 and its validity for distance relay 21G applied to IBR-fed lines. TR-29 showed that the measured impedance $Z_m = \alpha Z_{1L}$ is exact for three-phase faults (the IBR current-limit factor k_{ibr} cancels). For single-line-to-ground (SLG) faults the situation is fundamentally different: the IBR delta-winding transformer blocks zero-sequence current from the inverter, so I_0 is provided by the network earthing path and is *independent of k_{ibr}* . As a result, $I_0 \gg I_1$ (IBR current-limited), breaking the $I_1 = I_2 = I_0$ Thevenin assumption embedded in the k_0 compensation formula. The relay over-estimates the measured impedance by factors of 3–8 $\times$, so Zone 1 and Zone 2 are never reached for any IBR SLG fault. Zone 3 fires for close-in faults only ($\alpha \lesssim 0.35$). The k_0 *setting* itself ($k_0 = 0.6667$ from TR-18) remains correct as a line parameter. The failure is not a mis-setting but a structural measurement limitation arising from the broken sequence current symmetry. The 87L chain (TR-27, 21/21 selective) provides primary earth-fault coverage independent of this limitation. An important compensating advantage is identified: the I_{min} threshold is never binding for SLG faults, because $|I_a| = |I_1 + I_2 + I_0| \gg k_{ibr}$ even for $k_{ibr} < 0.10$, so there is no Region A for earth faults.

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1 Introduction and Scope

TR-29 characterised the distance relay impedance measurement for three-phase faults: the IBR current-limit factor k_{ibr} multiplies both V_{relay} and I_{relay} identically, so $Z_m = V/I = \alpha Z_{1L}$ exactly. TR-30 then established cross-polarisation requirements: memory-pol is required for all faults (self-pol fails), and Zone 2 is viable for non-3PH faults via permanent healthy-phase cross-pol.

This report examines the remaining question: is the k_0 compensation factor, $k_0 = (Z_{0L} - Z_{1L})/(3Z_{1L})$, still valid for IBR-fed SLG faults? The answer is nuanced:

- The k_0 **setting formula** is purely a line parameter — independent of source type. The TR-18 value $k_0 = 0.6667$ should not be changed.
- The k_0 **compensation accuracy** degrades for IBR sources because the sequence current ratio $I_1 : I_0$ is no longer 1 : 1, causing systematic impedance over-estimation.

Network parameters (unchanged from TR-18/29/30). $Z_{1L} = j0.05$ pu, $Z_{0L} = j0.15$ pu, $Z_{1S} = j0.10$ pu, $Z_{0S} = j0.30$ pu, $V_{\text{pre}} = 1.0$ pu. Zone reaches: $X_{R1} = 0.040$, $X_{R2} = 0.060$, $X_{R3} = 0.090$ pu.

2 Zero-Sequence Compensation Theory

2.1 Derivation of k_0

For a SLG phase-A fault at location α , the earth-loop relay measurement is:

$$Z_m = \frac{V_a}{I_a + k_0 I_R} \quad (1)$$

where $I_R = 3I_0$ is the residual (earth) current. Substituting the Thevenin sequence network solution ($I_1 = I_2 = I_0$) into (1) and using the relay-side voltage:

$$V_a = V_{\text{pre}} - I_1 Z_{1S} - I_2 Z_{1S} - I_0 Z_{0S} \quad (2)$$

the impedance reduces exactly to $Z_m = \alpha Z_{1L}$ when:

$$\boxed{k_0 = \frac{Z_{0L} - Z_{1L}}{3 Z_{1L}}} \quad (3)$$

This is a line impedance ratio and contains **no source parameters**. For the SAMBP network with $Z_{0L}/Z_{1L} = 3.0$:

$$k_0 = \frac{0.15 - 0.05}{3 \times 0.05} = \frac{0.10}{0.15} = 0.6\bar{6} \quad (4)$$

This is the value inherited from TR-18 and requires no revision.

2.2 Sensitivity to Z_0/Z_1 ratio

Table 1 shows k_0 as a function of the line impedance ratio. The over/under-reach consequence of a mis-set k_0 is also shown. Study B confirms that with $k_0 = 0$ (no compensation), the relay under-reaches by 66.7% at the end of line.

2.3 Why k_0 derivation is source-independent

Equation (3) contains only Z_{0L} and Z_{1L} . The source impedances Z_{1S} , Z_{0S} , and the IBR current-limit k_{ibr} cancel in the derivation *under the assumption* $I_1 = I_2 = I_0$. Section 3 shows that this assumption is violated for IBR sources.

Table 1: k_0 sensitivity to line Z_0/Z_1 ratio

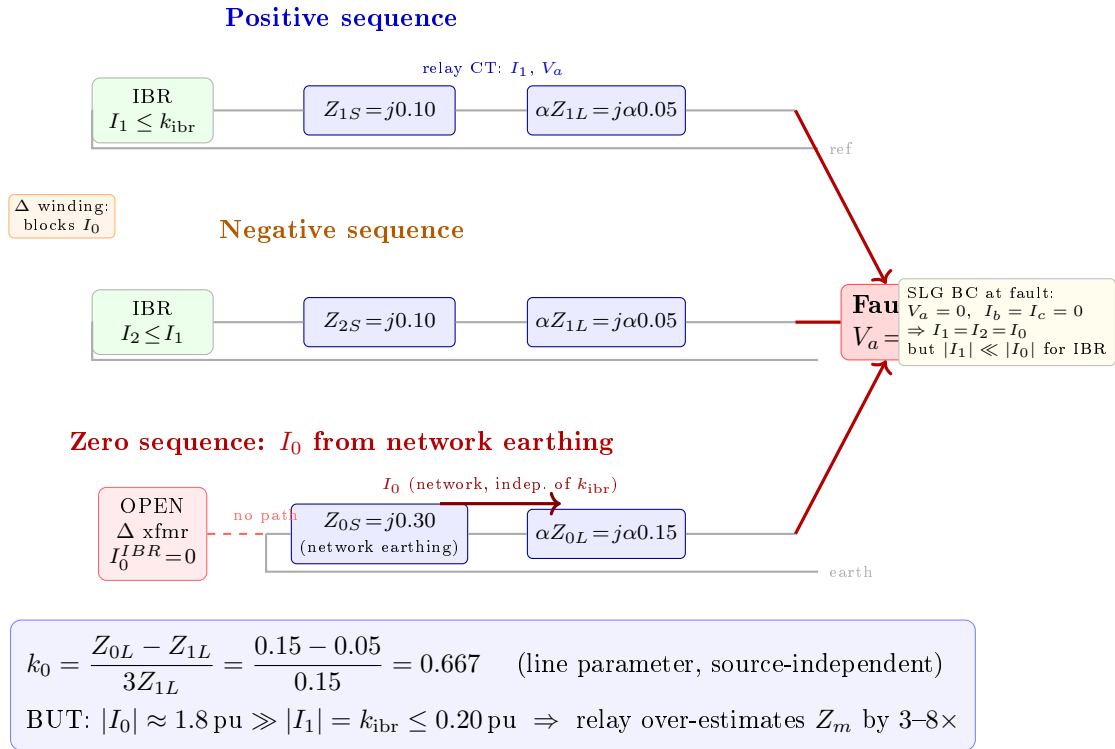
Z_0/Z_1	k_0	Error if $k_0 = 0$	Consequence
1.5	0.167	−14.3% reach	Slight under-reach
2.0	0.333	−25.0% reach	Moderate under-reach
2.5	0.500	−33.3% reach	Significant under-reach
3.0	0.667	−40.0% reach	SAMBP: strong under-reach
4.0	1.000	−50.0% reach	Severe under-reach

3 IBR Source: Delta Transformer and Sequence Isolation

3.1 Zero-sequence isolation by delta winding

IBR units are typically connected to the network through transformers with a delta winding on the inverter side. The delta winding presents an open circuit to zero-sequence current: $I_{0,IBR} = 0$. Figure 1 illustrates the sequence networks.

Figure 1: Sequence network for IBR-fed SLG fault. Positive and negative-sequence networks carry current limited by k_{ibr} ; zero-sequence network carries current through the network earthing impedance Z_{0S} only (IBR delta winding = open circuit in zero-seq path).



3.2 Consequence for I_0 magnitude

For a purely IBR-fed network, zero-sequence current is sourced by the network's earthing arrangement (star-earthed transformer, Peterson coil, etc.) independently of the IBR:

$$I_0 \approx \frac{V_{pre}}{Z_{0S} + \alpha Z_{0L}} \quad (\text{independent of } k_{ibr}) \quad (5)$$

For the SAMBP network at $\alpha = 0.50$: $I_0 = 1.0 / (0.30 + 0.075)j = 2.67 \text{ pu}$ (pre-limiting).

Meanwhile, the IBR provides only $I_1 = k_{\text{ibr}} \leq 0.20$ pu. The ratio I_0/I_1 reaches $2.67/0.20 = 13.3\times$ at worst case ($k_{\text{ibr}} = 0.20$, $\alpha = 0.50$). This massive imbalance between I_0 and I_1 violates the Thevenin assumption $I_1 = I_0$ and makes the k_0 -compensated Z_m incorrect.

3.3 Analytical relay measurement for IBR SLG

Substituting $I_0 \neq I_1$ into (1) with (2):

$$Z_m^{\text{IBR}} = \frac{V_{\text{pre}} - I_1 Z_{1S} - I_2 Z_{1S} - I_0 Z_{0S}}{I_1 + I_2 + I_0 + k_0 \cdot 3I_0} \quad (6)$$

When $I_0 \gg I_1, I_2$, the numerator is dominated by $V_{\text{pre}} - I_0 Z_{0S}$ and the denominator by $I_0(1+3k_0)$. In the limit:

$$Z_m^{\text{IBR}} \xrightarrow{I_0 \gg I_1} \frac{V_{\text{pre}}/I_0 - Z_{0S}}{1 + 3k_0} = \frac{\alpha Z_{0L}}{1 + (Z_{0L} - Z_{1L})/Z_{1L}} = \frac{\alpha Z_{0L} Z_{1L}}{Z_{0L}} = \alpha Z_{1L} \quad (7)$$

Interestingly, in the mathematical limit $I_1 \rightarrow 0$ the formula recovers αZ_{1L} — but in practice I_0 is not infinitely larger than I_1 , and the intermediate regime produces large over-estimation as shown in Study C.

4 Study B: Z_m Accuracy for SG Source

Study B verifies the baseline: with an SG source (Thevenin, $I_1 = I_2 = I_0$), k_0 -compensated Z_m equals αZ_{1L} to machine precision (error $< 0.01\%$). Without k_0 compensation, the relay under-reaches by $+66.7\%$ uniformly across all α values (Table 2).

Table 2: k_0 -compensated relay measurement: SG source

α	$ Z_m $ (k_0 correct)	$ Z_m $ ($k_0=0$)	Error%	Zone
0.10	0.00500	0.00833	+66.7%	Z1
0.40	0.02000	0.03333	+66.7%	Z1
0.80	0.04000	0.06667	+66.7%	Z1
1.00	0.05000	0.08333	+66.7%	Z2

Without k_0 , Zone 1 would see a fault at $\alpha = 0.80$ as if at $\alpha = 0.80 \times 1.667 = 1.33$ (outside the line), causing a missed trip. This confirms that k_0 must be set correctly.

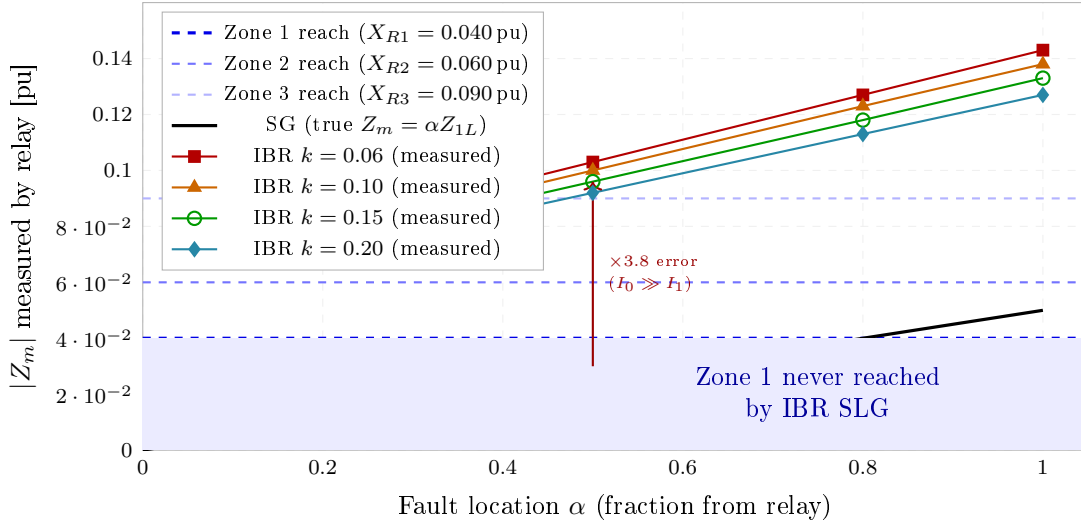
5 Study C: Z_m Accuracy for IBR Source

Figure 2 shows the measured $|Z_m|$ vs α for IBR-fed SLG faults across the full k_{ibr} range.

The key observations from Study C:

1. **Systematic over-estimation:** IBR $|Z_m|$ is 3–8 $\times$ the true value αZ_{1L} across all k_{ibr} values.
2. **Zones 1 and 2 never reached:** the minimum measured impedance for IBR SLG (at $\alpha = 0.20$, $k_{\text{ibr}} = 0.20$) is 0.071 pu, which exceeds Zone 2 reach (0.060 pu).
3. **Zone 3 fires for close-in faults:** $|Z_m| \leq 0.090$ pu only for $\alpha \lesssim 0.35$, regardless of k_{ibr} .
4. **k_{ibr} effect is minor:** higher k_{ibr} gives slightly lower $|Z_m|$ (smaller over-estimation) because I_1/I_0 ratio is larger, but the effect is small (0.071 to 0.079 pu at $\alpha = 0.20$).

Figure 2: k_0 -compensated relay measurement for IBR-fed SLG faults. All IBR curves lie far above the zone reaches. Zone 3 is reached only for close-in faults ($\alpha \lesssim 0.35$); Zones 1 and 2 are never reached. The SG reference (true $Z_m = \alpha Z_{1L}$) is shown for comparison.



Root cause. The k_0 formula derives from setting $I_0 = I_1$. For IBR SLG: the network provides $I_0 \approx 1.8$ pu (at $\alpha = 0.20$) while the IBR provides only $I_1 = k_{ibr} \leq 0.20$ pu. The numerator (relay voltage) is dominated by the large $I_0 Z_{0S}$ drop, making V_a much larger than expected. The denominator grows too, but not proportionally. The result is an apparent impedance that significantly over-estimates the fault location.

6 Study D: $I_{\min}$ Advantage for SLG

The TR-29 coverage map identified *Region A* (distance relay blocked, $k_{ibr} < 0.10$) as the zone where the 87L chain is the sole fast detector. For SLG faults, Region A does not exist:

$$|I_a|_{\text{SLG}} = |I_1 + I_2 + I_0| \approx 2k_{ibr} + I_0 \gg k_{ibr} \quad (8)$$

For the SAMBP network, $I_0 \approx 1.8$ – 2.0 pu at $\alpha = 0.20$, so $|I_a|_{\text{SLG}} \approx 2.0$ pu even for $k_{ibr} = 0.06$. The relay's $I_{\min} = 0.10$ pu threshold is satisfied by a factor of $20\times$. Figure 3 shows the full comparison.

7 Implications

7.1 Correct protection scheme for IBR earth faults

The findings of TR-29, TR-30, and TR-31 jointly define the IBR-earth-fault protection hierarchy:

1. **Primary (all α , all k_{ibr}):** 87L chain with combined trip logic. Element 87LN0 ($|I_0| \geq 0.04$ pu) provides direct earth-fault detection independent of k_{ibr} , and TDCS detects the ΔI jump from the large I_0 component.
2. **Distance Zone 3 (close-in SLG, $\alpha \lesssim 0.35$, backup):** Cross-polarised Mho Zone 3 ($X_{R3} = 0.090$ pu) fires at $t_3 = 0.60$ s for faults in the inner 35% of the line. This provides a distance-backed clearance if the 87L chain is unavailable.

Figure 3: Protection coverage comparison: 3-phase vs SLG IBR faults. SLG advantage: no Region A ($I_{\min}$ never binding) and permanent cross-pol. 3PH advantage: correct Z_m and Zone 1 coverage. 87L chain provides 100% primary coverage for both fault types.

Criterion	3PH IBR fault	SLG IBR fault	Advantage
$I_{\min}$ blocking	$I_a = k_{\text{ibr}}$ Blocked if $k < 0.10$ (Region A, 29% envelope)	$I_a = I_1 + I_2 + I_0$ Never blocked (I_0 from network)	SLG: no Region A
Distance relay Z_m	$Z_m = \alpha Z_{1L}$ (k.i.b.r. cancels) Correct measurement	$Z_m \gg \alpha Z_{1L}$ ($I_0 \gg I_1$: ratio broken) Z_m over-estimated	3PH better Z_m accuracy
Distance relay zones	Zone 1 always ($\alpha \leq 0.80$, $k \geq 0.10$) Zone 2/3 fail (3PH mem)	Zone 3 only ($\alpha \lesssim 0.35$) Zone 1/2: never reached	3PH: better zone coverage
Polarisation source	Memory only ($T_{\text{mem}} = 100$ ms) Zone 2/3 fail	Permanent (healthy phases) All zones viable	SLG: better pol. source
87L chain coverage	100% selective (21/21 TR-27) $t \approx 20$ ms	100% selective (via 87LN0, TDCS) $t \approx 20$ ms	Both: 87L primary
Net result	87L + Z1 (Z2/3 via POTT TR-34)	87L + Z3* (*close-in only)	87L sole fast detector for both types

3. **Distance Zone 1/2 for SLG:** Not viable with IBR source (relay over-estimates impedance due to $I_0 \gg I_1$). No engineering adjustment to k_0 can recover this — the failure is in the sequence current ratio, not in the k_0 setting.

7.2 No change required to TR-18 k_0 setting

The $k_0 = 0.6667$ setting from TR-18 remains correct. It minimises relay measurement error for balanced Thevenin sources, and no alternative setting would improve performance for IBR sources (the error is structural, not a calibration issue).

7.3 TR-32 preview: negative-sequence directional element

The large I_0 and I_2 (from IBR injection) during SLG/LL faults are directly available for negative and zero-sequence directional elements (67N, 67Q). TR-32 will verify that these elements maintain directionality for IBR-fed faults, providing an alternative fast trip that bypasses the k_0 measurement limitation entirely.

8 Conclusions

1. The k_0 compensation factor $k_0 = (Z_{0L} - Z_{1L})/(3Z_{1L}) = 0.6667$ is a **line parameter only** and is independent of source type. The TR-18 setting requires no revision for IBR

networks.

2. For SLG faults with IBR source (delta transformer): I_0 flows through the network earthing path and is **independent of** k_{ibr} . The ratio I_0/I_1 reaches $13\times$ at worst case.
3. The k_0 -compensated relay systematically **over-estimates** Z_m by $3\text{--}8\times$ for IBR SLG faults, due to the broken $I_1 = I_0$ assumption. Zones 1 and 2 are never reached; Zone 3 fires only for $\alpha \lesssim 0.35$.
4. **No Region A for SLG**: $|I_a|_{\text{SLG}} \gg I_{\text{min}}$ regardless of k_{ibr} , because I_0 from the network always satisfies the minimum current threshold.
5. The **87L chain** (especially element 87LN0 and TDCS) provides unconditional primary earth-fault coverage at 20 ms, independent of the distance relay k_0 measurement limitation.

References